similar target. Therefore, the client does not need to know about the target; it knows only about the load balancer.

Asynchronous decoupling is different and commonly used in scenarios where the client does not expect an immediate response. For example, a web application could scale and optimize an image uploaded by the user in the background and use the raw image until that process finished in the background. For asynchronous decoupling, AWS offers the *Simple Queue Service* (SQS), which provides a message queue. The producer sends a message to the queue, and a receiver fetches the message from the queue and processes the request.

You'll learn about both the ELB and the SQS services in this chapter. Let's start with ELB.

## *14.1 Synchronous decoupling with load balancers*

Exposing a single EC2 instance running a web server to the outside world introduces a dependency: your users now depend on the public IP address of the EC2 instance. As soon as you distribute the public IP address to your users, you can't change it anymore. You're faced with the following problems:

- Changing the public IP address is no longer possible because many clients rely on it.
- If you add an additional EC2 instance (and IP address) to handle the increasing load, it's ignored by all current clients: they're still sending all requests to the public IP address of the first server.